OpenADMET PXR Methodology Report

Environment

Software
: Python in dedicated Conda environments under Windows. Code generated using Google Gemini.

Hardware
: Very limited. CPU only, Up to 12Gb available memory (optimistically) out of 32Gb installed. Attempts to use the laptop GPU met with limited success and frequent crashes.

Method - Regression

The training script implements a **Chemprop v2 hybrid architecture** optimized for regression using multi-modal feature fusion.

1. Things that helped

- Multitarget models rather than individual models
- Restrict the training dataset to an explicit molecular weight range of 160 - 600.
- Augment the graph representation with the Chemeleon fingerprints
- Weight rows by number of occurrences in that Cyp
- Apply affine transformation to 2D6 predictions (as per Brian Wylie)
- Dropout = 0.2
- CV by structure (Murcko-Bemis scaffolds)

2. Things that didn’t help

- No dropout
- Cross validate by average activity over all Cyps
- Cross validate by molecular weight
- Use ORGANIC atom featurizatio
- Overweight 2D6 rows
- Uncertainty Quantification via Evidential Loss
- Freeze MPNN for first 10 epochs, then unconstrain, with early stopping (patience = 10 epochs).
- Hybrid quantile and direct margin ranking loss WITH affine transformation

Method - Classification

Classification was performed using chemprop and chameleon descriptors using Autogluon classification and the following parameters

- Best preset mode